

ROC CURVE / AUC FOR LOGISTIC REGRESSION

To evaluate the discriminatory performance of the logistic regression model in distinguishing churned from non-churned customers. By comparing the tenure, monthly charges, total charges, contract, internet service, technical support and payment method.

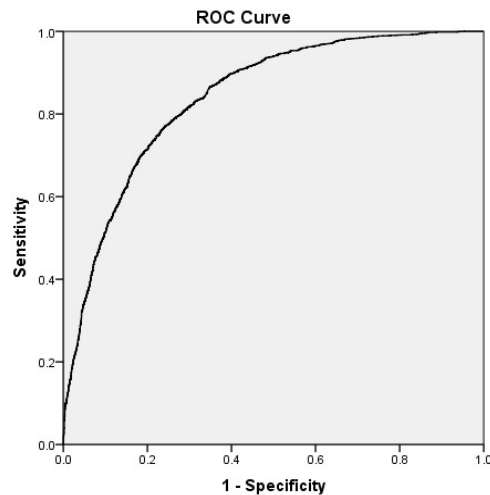
RESULT

Case Processing Summary

churn_num	Valid N (listwise)
Positive ^a	1869
Negative	5163
Missing	11

Larger values of the test result variable(s) indicate stronger evidence for a positive actual state.

a. The positive actual state is 1.



Area Under the Curve

Test Result Variable(s):
Predicted probability

Area
.840

The test result variable(s): Predicted probability has at least one tie between the positive actual state group and the negative actual state group. Statistics may be biased.

INTERPRETATION

The logistic regression model achieved an AUC of 0.840, indicating good discriminatory performance in distinguishing churned from non-churned customers.